Essentials of Data Science Lab Assignments

Course Code: 2304102L | AY: 2023-2024

Practical No. 1: Python Fundamentals

Practice Lab Assignments

1. Accept object mass (kg) and velocity (m/s) and display momentum ($e=mc^2$).

```
mass = float(input('Enter mass (kg): '))
velocity = float(input('Enter velocity (m/s): '))
# Note: Syllabus mentions e=mc^2 for momentum, though physically p=mv.
# We will follow the syllabus formula provided.
momentum = mass * (velocity ** 2)
print(f'Momentum: {momentum}')
```

2. Print square (1-digit), square root (2-digit), or cube root (3-digit) of n.

```
import math
n = int(input('Enter a number: '))
if 0 <= n <= 9:
    print(f'Square: {n**2}')
elif 10 <= n <= 99:
    print(f'Square root: {math.sqrt(n)}')
elif 100 <= n <= 999:
    print(f'Cube root: {n**(1/3)}')
else:
    print('Number outside specified range.')
```

3. Transform birthdate to age and salary (INR to USD).

```
from datetime import date

def transform_data(birth_date, salary_inr):
    today = date.today()
    age = today.year - birth_date.year - ((today.month, today.day) <
(birth_date.month, birth_date.day))
    salary_usd = salary_inr / 83.0  # Approx conversion rate
    return age, salary_usd

# Example usage
b_date = date(1995, 5, 20)
sal_inr = 500000
age, usd = transform_data(b_date, sal_inr)
print(f'Age: {age}, Salary (USD): ${usd:.2f}')
```

4. Print the reverse of a given number.

```
num = int(input('Enter a number: '))
reverse = int(str(num)[::-1])
print(f'Reversed number: {reverse}')
```

Lab Assignments

1. Compute result for 5 courses (Pass >= 40, Distinction > 75%, etc).

```
marks = [float(input(f'Course {i+1} marks: ')) for i in range(5)]
if any(m < 40 for m in marks):
    print('Result: Fail')
else:
    avg = sum(marks) / 5
    if avg > 75: grade = 'Distinction'
    elif avg >= 60: grade = 'First Division'
    elif avg >= 50: grade = 'Second Division'
    elif avg >= 40: grade = 'Third Division'
    print(f'Average: {avg}%, Grade: {grade}')
```

2. Fibonacci sequence using recursion.

```
def fibonacci(n):
    if n <= 1: return n
    return fibonacci(n-1) + fibonacci(n-2)

terms = 10
print([fibonacci(i) for i in range(terms)])
```

3. Print different patterns.

```
rows = 5
for i in range(1, rows + 1):
    print('* ' * i)
```

Practical No. 2: Data Structures (List & Dictionary)

Practice Lab Assignments

1. Perform all List and Dictionary operations.

```
my_list = [1, 2, 3]; my_list.append(4); my_list.remove(2)
my_dict = {'a': 1, 'b': 2}; my_dict['c'] = 3; del my_dict['a']
```

Lab Assignments

1. Linear Search to find position in a list.

```
def linear_search(arr, x):
    for i in range(len(arr)):
        if arr[i] == x: return i
    return -1

nums = [10, 20, 30, 40]
print(f'Position: {linear_search(nums, 30)}')
```

2. Find cricket team captain (tallest person) using a dictionary.

```
team = {'Player1': 180, 'Player2': 192, 'Player3': 175}
captain = max(team, key=team.get)
print(f'Captain: {captain} (Height: {team[captain]}cm)')
```

Practical No. 3: NumPy Operations

Practice Lab Assignments

1. Perform all NumPy operations.

```
import numpy as np
arr = np.array([1, 2, 3])
print(f'Mean: {np.mean(arr)}, Sum: {np.sum(arr)}')
```

Lab Assignments

1. Matrix operations, stacking, slicing, and statistical operations on real-life data.

```
import numpy as np
data = np.array([[10, 20], [30, 40]])
# Matrix Ops
print(f'Transpose:\n{data.T}')
# Stacking
print(f'Vertical Stack:\n{np.vstack((data, data))}')
# Statistics
print(f'Column-wise Sum: {np.sum(data, axis=0)}')
```

Practical No. 4: Pandas for Data Manipulation

Practice Lab Assignments

1. Perform all Pandas operations.

```
import pandas as pd
df = pd.DataFrame({'A': [1, 2], 'B': [3, 4]})
print(df.describe())
```

Lab Assignments

1. Analyze Sales Dataset (Best month, top product, top city).

```
import pandas as pd
# Mocking a Sales Dataset
data = {'Month': ['Jan', 'Feb', 'Jan'], 'Sales': [200, 150, 300], 'Product':
['A', 'B', 'A'], 'City': ['NY', 'LA', 'NY']}
df = pd.DataFrame(data)

# Best month
best_month = df.groupby('Month')['Sales'].sum().idxmax()
# Top Product
top_prod = df['Product'].value_counts().idxmax()

print(f'Best Month: {best_month}')
print(f'Top Product: {top_prod}')
```

Practical No. 5: Data Visualization (Matplotlib)

Practice Lab Assignments

1. Install Matplotlib and draw basic graphs.

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [4, 5, 6])
plt.title('Basic Plot')
plt.show()
```

Lab Assignments

1. Titanic Dataset analysis and visualization (5 grains/graphs).

```
import matplotlib.pyplot as plt
import seaborn as sns
# Using built-in titanic dataset for demonstration
titanic = sns.load_dataset('titanic')

# Grain 1: Survival Count
titanic['survived'].value_counts().plot(kind='bar')
plt.title('Survival Count')
plt.show()
```